Reg. No. | | | | | | | | | | |

[22BS1MA02]

FIRST SEMESTER B.Tech EXAMINATION, DECEMBER - 2022

COMPUTATIONAL MATHEMATICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

PART-A (5×2=10 Marks)

ANSWER ALL QUESTIONS

1	a.	Find the cardinality of the given set : $S = \{x: x^2 - 16 = 0\}$.	L2	CO1	02
	b.	Find the power set A where $A = \{1,2,3\}$.	L3	CO2	02
	c.	Construct the truth table for the given propositions: (p ∨ ¬q).	L3	CO3	02
	d.	Find the rank of a matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$.	L3	CO4	02
	e.	Solve the differential equation $4y'' + 4y' - 3y = 0$.	L3	CO5	02

PART - B (6×5=30)

2.	a.	Discuss the python code for a set which contains elements from set A but not from set B, where $A = \{x: x \text{ is a prime number} < 15\}$ and $B = \{x: x \text{ is an odd number} < 15\}$.	L2	CO1	05
		OR			
	b.	Identify the frozen set in python which contains elements from set A and set B but not both, where A and B are frozen sets.	L2	CO1	05
3.	a.	Discuss python code to find Eigen values and the corresponding Eigen vectors and print the same of the matrix $\begin{bmatrix} 2 & 4 \\ 1 & -3 \end{bmatrix}$.	L3	CO2	05
		OR			
	b.	Discuss python code to find Eigen values and the corresponding Eigen vectors and print the same of the matrix $\begin{bmatrix} 2 & -3 & 0 \\ 2 & -5 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.	L3	CO2	05
4.	a.	Use truth table to identify whether the compound proposition $p \to (p \vee q)$ is a tautology and the compound proposition $p \wedge (\neg p \wedge q)$ is a contradiction.	L3	CO3	05
		OR			
	b.	Use rules of inference to check the validity of the following argument: If Ravi goes out with friends, he will not study. If Ravi does not study, his father becomes angry His father is not angry ∴ Ravi has not gone out with friends	L3	CO3	05

5.	a.	Solve the system $X' = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} X$ by diagonalization method.	L3	CO4	05
		OR			
	b.	Test the consistency and hence solve the system of equations : $5x + 3y + 7z = 5$, $3x + 26y + 2z = 9$ and $7x + 2y + 10z = 5$.	L3	CO4	05
6.	a.	Solve the differential equation $y''' - 5y'' + 7y' - 3y = 0$.	L3	CO5	05
		OR			
	b.	Solve the differential equation $(D^2 - 2D + 2)y = \cos(x - 1)$.	L3	CO5	05
7.	a.	Find the directional derivative of the function $f = x^2yz + 4xz^2$ at the point (1,-2,-1) along the direction of the vector $2\hat{\imath} - \hat{\jmath} - 2\hat{k}$.	L4	CO6	05
		OR			
	b.	Examine whether the function $f(x,y) = x^3 + y^3 - 3xy + 1$ is minimum at the point (1,1).	L4	CO6	05

PART-C (6X10=60 Marks)

8.	a.	I. Let A and B be finite sets with $\lvert B \rvert = 3$. If there are 4096 relations from A to B then identify the cardinality of A. II. If $A=\{1,2,3\}$ and $B=\{p,q,r\}$, then find $A\times B$, $B\times A$ and $\lvert A\times B \rvert$.	L3	CO2	10
		OR			
	b.	Explain that the relation congruence modulo m, $a \equiv b \pmod{m}$ on the set of all positive integers Z is an equivalence relation.	L2	CO1	10
9.	a.	Find Eigen values and the corresponding Eigen vectors of the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$.	L3	CO2	10
		OR			
	b.	Diagonalize the matrix $A = \begin{bmatrix} 11 & -4 & -7 \\ 7 & -2 & -5 \\ 10 & -4 & -6 \end{bmatrix}$.	L3	CO2	10
10.	a.	Use python code to interpret the following compound proposition: i. $p \wedge (\neg p \wedge q)$ ii. $(p \vee q) \rightarrow r$	L3	CO3	10
		OR			
	b.	Explain whether the following is a valid argument by using Python code: *If I study, then I do not fail in the exam.* *If I do not fail in the exam, then my father gifts a car to me.* *∴ If I study then my father gifts me a car.*	L3	CO3	10
11.	a.	Solve the differential equation $(D^4 - 18D^2 + 18)y = 36e^{3x} + 8^x$.	L3	CO4	10
		OR			
	b.	Check for consistency and hence solve the system of equations $5x_1 + x_2 + 3x_3 = 20$, $2x_1 + 5x_2 + 2x_3 = 18$ and $3x_1 + 2x_2 + x_3 = 14$.	L3	CO4	10

12.	**a.**	Solve the differential equation $(D^2+4)y = 4\sin 2x + \cos 5x$.	L3	CO5	**10**
		OR			
	b.	Solve the initial value problem $(D^3 - D^2 - 4D + 4)y = 0$, where $y(0) = 1$, $y'(0) = 2$ and $y''(0) = 1$.	L3	CO5	**10**
13.	**a.**	Define solenoidal and irrotational and then test whether $\vec{F} = (y^2 - z^2 + 3yz - 2x)\hat{\imath} + (3xz + 2xy)\hat{\jmath} + (3xy - 2xz + 2z)\hat{k}$ is both solenoidal and irrotational.	L4	CO6	**10**
		OR			
	b.	Show that the total derivative $\frac{du}{dt} = -\frac{2}{e^{2t}+e^{-2t}}$, where $u = tan^{-1}\left(\frac{y}{x}\right)$, $x = e^t - e^{-t}$ and $y = e^t + e^{-t}$.	L4	CO6	**10**